

Job Description

Section 1 Post Details	
Job Title:	Senior Data Engineer
Pay Band:	8a
Section 2 Post Details	
Division:	Corporate
Department:	Digital
Accountable to:	Chief Digital and Information Officer (CIO)
Reports to:	Head of Business Intelligence
Job Summary:	The Senior Data Engineer will lead the development and on-going maintenance of technologies that facilitate researchers' safe and secure access to health datasets for the Trust. This is a critical role supporting both operational excellence and research innovation, enabling the organisation to harness data effectively to improve patient outcomes and meet regulatory obligations. The Senior Data Engineer will play a pivotal role in designing, developing, and maintaining modern data infrastructure to support the analytical and reporting needs of The Clatterbridge Cancer Centre. This role will focus heavily on the Microsoft BI stack, with particular emphasis on Azure Fabric, to build and manage a scalable, secure, and high-performing data warehouse within a Secure Data Environment (SDE). The Trust has invested in Microsoft Azure platform with an ambition to provide a highly functional research facility and therefore a significant component of the role involves



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	contributing to the design and delivery of a Trusted Research Environment (TRE). This will require a strong understanding of secure data access frameworks, governance and compliance, with particular attention to the handling of sensitive healthcare related data and the complexities this involves. The role holder will be expected to champion best practices in data engineering, including CI/CD, version control, testing, and documentation. They will also collaborate with Research & Innovation, academic institutes, pharmaceutical companies, data analysts, BI developers and other stakeholders to ensure alignment with enterprise architecture, data strategy and governance.
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Section 3 Key Responsibilities	
1. Key Responsibilities	Responsibility for Information Resources 1. Develop, construct, test and maintain data and processing workflows to architectural standards. 2. Build robust, efficient and reliable data pipelines 3. Integrating external or new datasets into existing data pipelines 4. Develop solutions for data acquisition 5. Develop dataset processes for data modelling, mining, and production 6. Drive the collection of new data and refinement of existing data sources 7. Establish complex enterprise-scale data ingestion processes using standard development techniques. 8. Manage people and technical resources to ensure that data services work effectively at scale. 9. Oversee the operational implementation of new technologies and contribute to the modernisation of the business data architecture.



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	10. Governance requirements and develop ethical and effective data sharing agreements for data sources. 11. Contribute to organisational policies, standards, and guidelines for data engineering. 12. Co-ordinate the application of data design and modelling techniques, to establish, modify or maintain data pipelines, data structures, datasets, and metadata for diverse datatypes. 13. Work closely with the database administrators and integration team to ensure development and maintenance of the Trust Data Model(s) and associated architecture artefacts and technical standards. 14. Proactively manage the integrity of data: investigating, raising and resolving issues with the appropriate Trust managers. 15. Ensure that all reporting is fully compliant with data protection legislation, Trust policy and more generally that our reporting conforms to information governance best practice. Analytical and Judgement Skills 1. Lead the development of the data environment to support data for research. 2. Working with the wider business intelligence team to exemplify best practice with methods and tools and guide others to a high standard. 3. Produce timely and accurate briefing reports and other reports, on own initiative and on request as required. This involves analysing complex data and information from a range of sources. 4. Responsible for analysis of highly complex and sensitive data within the data warehouse and the investigation of new data sources. 5. Maintain a current working knowledge of NHS and Trust information systems, analytical techniques, departmental policies and procedures as required. Alert and advise other members of the team to information system
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	changes that may affect the recording or reporting of service performance. Planning and Organisational Skills 1. To develop and ensure implementation/compliance of strategies to enable the effective functioning of the Trust in a changing environment, responding to both internal and external pressures in line with, and when appropriate, influencing the Trust Digital Strategy. 2. Responsible for data mapping, detailed data analysis and metadata analysis; creating and maintaining mapping documents and technical documentation. To be responsible for generating the standard operating policies and procedures for the Trust's data warehouse and for updating these policies as developments are made 3. Contribute to the strategic planning of team projects, identifying interdependencies, and potential impacts on wider organisation, resource requirements and building in contingency and adjustments as necessary. 4. Contribute to short-, medium- and long-term business plans. Responsibility for Patient Care 1. Meeting the needs of clinical and managerial staff by providing appropriate information and business intelligence. 2. To ensure that data engineering solutions are developed with patient outcomes in mind. 3. Incidental contact within the care environment. Responsibility for Policy / Service Development 1. Keep up to date with emerging technologies, tools, data sets and methodologies in appropriate areas. 2. To translate national standards to the local arena and advise on data collection methods.
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3. To support the Head of Business Intelligence with the delivery of the Data Strategy ensuring all objectives are met.
4. Contribute to the continued improvement and quality of the Business Intelligence Function.
5. Develop policies and procedures in own work function with an impact on the wider organisation, as required.

Responsibility for Financial and Physical Resources

1. Provide advice and support on management information related issues during the procurement of new and replacement information systems.
2. Maintain technical reference documentation to support contingency, handover, maintenance and training.
3. Correct use and security of own resources within the workplace
4. Support regular reporting on the targeting of resources and monitoring their implementation from a value for money perspective.
5. Ensure project expenditure is monitored accordingly.

Responsibility for Human Resources

1. Provide direction and management in relation to data relating to research ensuring the team meets its objective of helping colleagues improve patient care by providing accurate, relevant and timely information.
2. Lead, manage and develop team, undertaking appraisals, recruitment and personal record keeping, professional leadership and technical direction.
3. To adhere to the NHS Managers Code of Conduct and any other relevant professional codes of conduct at all times



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4. Uses specialist knowledge to train staff, usually at senior levels to use tools and systems to obtain and utilise information.

Responsibility for Research and Development

1. Develop TRE capabilities and create secure data pipelines that allow researchers to work with and analyse a diverse portfolio of datasets.
2. Participate with audits and surveys, facilitating work with researchers as necessary.
3. Use research and development skills to build concise and accurate business cases addressing complex business issues.

Freedom to Act

1. Works autonomously with significant discretion. Work on own initiative. Guided by principles and broad occupational policies.
2. Maintain confidentiality of all patient information according to local guidelines.

Partnership Working

1. Establish effective working relationships with stakeholders, both internally and externally, including University of Liverpool, NHS and other public sector partners, other research organisations, suppliers, SMEs etc. to assist in establishing changing demands and trends to ensure we provide an exemplar service.
2. Develop an effective working relationship with other Digital teams, enabling the sharing of knowledge and other NHS related issues.
3. Work with colleagues across the organisation and wider Health Economy teams to help formulate and meet their information requirements to support the achievement of the Trusts' strategic objectives.

Equality and Diversity



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	1. Ensure staff, patients and the general public are treated in a respectful and courteous manner, as well as the general duty to ensure compliance with policies on equality and diversity. This job description is not intended to be an exhaustive list of duties, but it aims to highlight the typical key responsibilities of the post. It may be reviewed from time to time to ensure that it relates to the job as then being performed, or to incorporate required changes. This process will be conducted in consultation with the post holder in line with the Trust Job Evaluation Policy and Process.
2. Key Working Relationships	Communication and Relationships 1. Communicate verbally and in writing with staff at all levels both internal and external to the organisation to promote best practice and support the organisation in making informed decisions by the provision of appropriate and relevant information in often challenging environments covering complex, contentious or sensitive topics, including but not limited to process re-design and complex reporting issues. 2. Highly developed persuasive and motivational skills to negotiate timescales with key information users in support of facilitating work streams between the Trust and Digital. 3. Be a visible, accessible and authoritative figure upon whom those using the service can rely. 4. Regular verbal presentation of information to the highest level of the organisation. 5. Represent the Digital Department at relevant internal and external meetings including deputising for the Head of Business Intelligence as appropriate.
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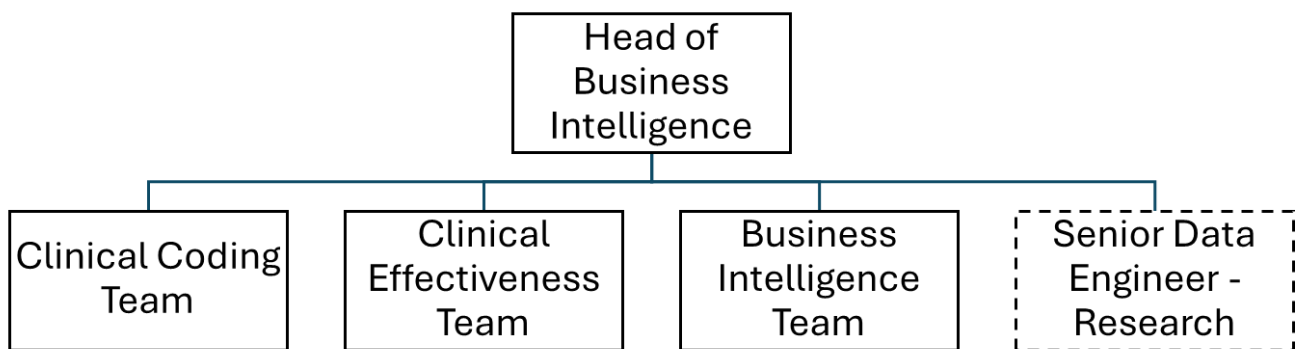


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Departmental Organisation Chart



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Section 4 Person Specification		
	Essential	Desirable
Qualifications:	Educated to degree level in a related subject or have significant relevant practical data engineering experience.	- Postgraduate level qualification in a relevant technical discipline or relevant equivalent experience, supported by specialist training and relevant study. - Membership and Professional Registration of a relevant Fed-IP recognised professional body is desirable (BCS, SOCITM, IHRIM, CILIP, AphA) - Formal Management / Leadership Training
Knowledge and expertise:	- An understanding of NHS Statutory Reporting, Data Flows and Commissioning Process - Ability to analyse and document business reporting requirements with an understanding of the resulting technical implications. - Current expert knowledge and experience of using SQL/ Synapse SQL, Python, Data Factory, Spark etc. - Understanding of the potential of data and the critical impact that quality data has across a complex organisation, a detailed understanding of data management practices.	Experience of working with a Patient Based Clinical Information System



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	• Able to formulate plans which cover a range of complex activities including target setting and prioritisation of activities. • Able to lead others, including the management of performance and delivery of outcomes, in line and non-line management relationships. • Commitment to continuing professional development.	
Experience:	• Experience of NHS data sets, broad understanding of NHS information procedures, legislation and policies. • Significant and relevant experience of working in the NHS or health setting. • Understanding of available data sources within an NHS Trust and how these can be extracted and combined for the benefit of the Trust. • Experience of requirements gathering and working with stakeholders to establish clear and realistic timeframes. • Experience of managing risk and reporting	• Knowledge of patient record systems and supporting procedures
Key skills and attributes:	• Understanding of data warehouse architecture including performance, maintainability and scalability along with building out a secure, governed and well architected data infrastructure.	• Microsoft Azure Certification



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	• Extensive knowledge with integration and transformation within Cloud Data Warehouses (e.g. Azure, AWS, GCP) • A good understanding of analytical (data science) model development and deployment lifecycle • Using data engineering expertise to architect the most appropriate solution design to suit a particular requirement or set of requirements • Selecting and using the most appropriate language and tools to develop data management and optimisation / data science solutions • Developing and overseeing delivery of test plans in line with Solution Assurance best practice • Applying expertise to the development of reference / master data management solutions • Strong communication skills at a technical and non-technical level to provide advice to colleagues on the most effective use of systems and information support to the Trust.	
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	• Demonstrable understanding of information infrastructure requirements in specialised and diverse organisations including data migration and data models. • Skilled negotiator able to develop relationships and manage stakeholders within IT, the business and IT third parties. • Able to concentrate intensively, using significant discretion in decision making activities. • Proactive and personally motivated • Self-aware, flexible and adaptable • Works effectively within a team environment. • Curious nature and effectively dives deep into material. • Expectation of travel across all areas of the Trust and Cheshire and Merseyside.	
Work related circumstances:	• Able to demonstrate the Trust core values and behaviours in day-to-day approach to work	



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Section 4

1. Mission & Values

Our Values support our Mission to drive improved outcomes and experience through our unique network of specialist cancer care across Cheshire & Merseyside.

Our Values represent who we are and what we believe in. They define how we act to deliver the best possible care for our patients and shape The Clatterbridge Cancer Centre as a great place to work.

We are:

- Kind
- Empowered
- Responsible
- Inclusive

2. Health and Safety

We will take reasonable practical steps to ensure your health, safety and wellbeing at work. You must familiarise yourself with our health and safety policies. It is your legal duty to take care for your own health and safety as well as that of others.

3. Infection Control

All employees are expected to follow consistently high standards of infection control practice, especially hand hygiene and adhere to dress/uniform code for clinical staff, aseptic technique and to be aware of and follow all infection control guidelines and procedures relevant to their work.

4. Equality and Diversity

All employees must demonstrate a positive attitude to our equality policies and Equality Scheme. Employees must not discriminate on the grounds of age, disability, gender reassignment, marriage and civil partnership, pregnancy and maternity, race, religion or belief, sex or sexual orientation and will treat patients, colleagues and members of the public with dignity and respect.

5. Competency of Health Professionals

To maintain professional registration (as appropriate) and to act in accordance with professional codes of practice and guidelines. To follow our policies and procedures and maintain up to date skills and knowledge through participation in continuing professional development.



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6. Partnership

To work in partnership to achieve Trust objectives and promote a culture of working together through good communications, openness and honesty.

7. Safeguarding

All employees have a responsibility to take appropriate action if they believe that a child or vulnerable adult is in need of services or in need of protection. You must be committed to safeguarding and promoting the welfare of children, young people and vulnerable adults. Everyone is responsible for accessing the relevant level of training and following the child protection and safeguarding procedures.

We comply with the requirements of the Disclosure and Barring Service (DBS) and the requirement to report safeguarding issues to the DBS. All staff required to have a DBS disclosure for their post will undergo a check every three years. Employees must cooperate with the renewal process and submit their DBS applications promptly when requested.

8. Mandatory Training

Mandatory training relates to information and/or training regarding the management of general and specific risk. All staff are required to attend mandatory training which is relevant to their role as identified in our risk management mandatory training matrix.

9. Data Protection, Information Security and Confidentiality

We have several policies relating to data protection, information security and a Code of Conduct for the handling of person identifiable information. All staff must protect our information, information assets, systems and infrastructure. During the course of your employment you may have access to, see or hear information of a confidential nature. You are required not to disclose such information, particularly relating to patients or staff. All person identifiable information must be held in the strictest confidence and should only be disclosed to authorised people in accordance with NHS confidentiality guidelines (Caldicott) and the Data Protection Act 2018, unless explicit written consent is given by the person identified or where information sharing protocols exist. Any failure to comply with this term of your employment will be treated as an act of misconduct under the Trust Disciplinary Policy. Staff must maintain their knowledge of the principles of data protection and information security.

10. Code of Conduct

As an employee of The Clatterbridge Cancer Centre, it is a contractual duty that you abide by any relevant code of professional conduct and/or practice applicable to you. A breach of this requirement may result in action being taken against you (in accordance with the Disciplinary Policy) up to and including dismissal. In addition, managers are required to carry out their duties in a manner which complies with the code of conduct for NHS Managers Directions 2002.



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11. Research Governance

Research and Development is at the heart of providing effective treatments and high quality services, supporting a culture of evidence based practice and innovation amongst staff. All staff have a duty to be aware of and comply with their responsibilities for research governance, whether as researchers, as part of the team caring for those participating in research, or as research participants themselves.

12. Smoking

Our hospitals are smoke-free. Smoking (including e-cigarettes) is not permitted anywhere in our buildings, including doorways and external grounds. Employees are not permitted to leave our sites to smoke during their contractual hours except during localised agreed breaks. If staff smoke off site, uniforms and any organisational identification must be fully covered to ensure that they cannot be recognised as The Clatterbridge Cancer Centre staff.

13. Travel Requirements

In line with business needs and requirements of the post, you must be able to demonstrate potential travel requirements.

14. On-call

The post may require you to participate in an on-call rota. In these circumstances your line manager will inform you of the pattern and regularity of this commitment and reimbursement will be in line with our terms and conditions.

Additional information for Job Matching

Physical Skills:

- Required to undertake prolonged periods of concentration



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	• Daily use, often for prolonged periods, of personal computer • Advanced keyboard skills to facilitate the manipulation of complex data at speed
Physical Effort:	• Office based and required to visit staff/ managers across all departments, including visiting CCC sites and other NHS Trust sites.
Mental Effort:	• Required to undertake prolonged periods of concentration
Emotional Effort:	• Often required to work in situations of high emotional tension and to deal with members of staff who may be angry or distressed
Working Conditions:	• Office based



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